PCBC H Name _____ _____ /22

## Limits Quiz 1.1–1.2

*Non-Calculator*

You have 20 minutes.

**PRACTICE C · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator**

**Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.**

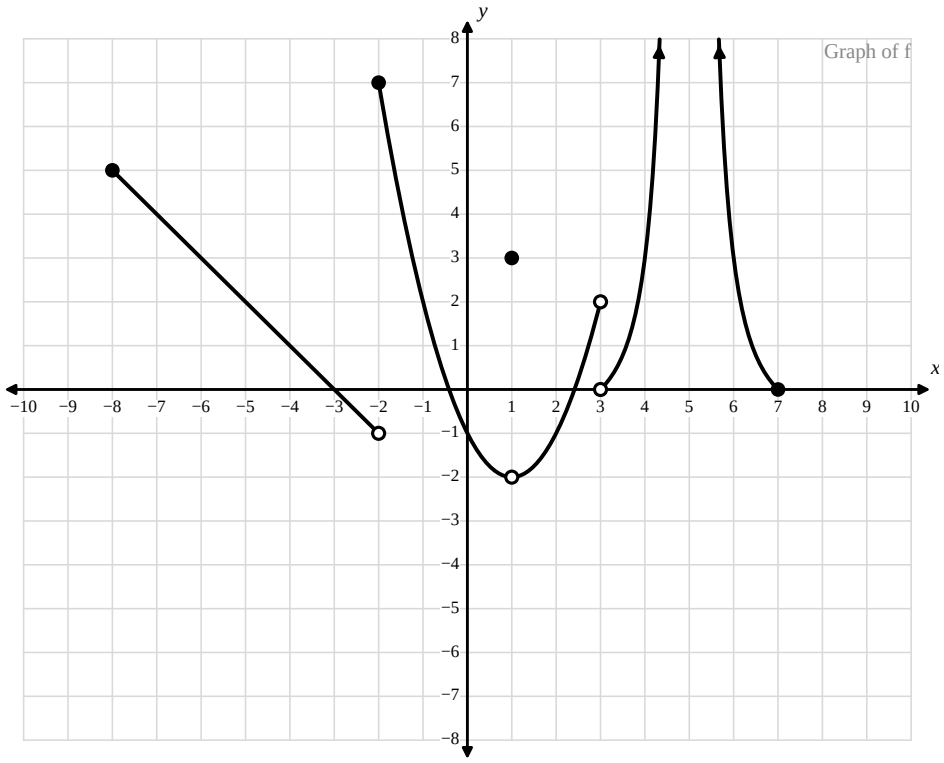
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Signature _____

**SHOW ALL WORK FOR FULL CREDIT.**

**Use the graph of $f$ shown below to find each. If a limit does not exist, write DNE. (2 points each)**

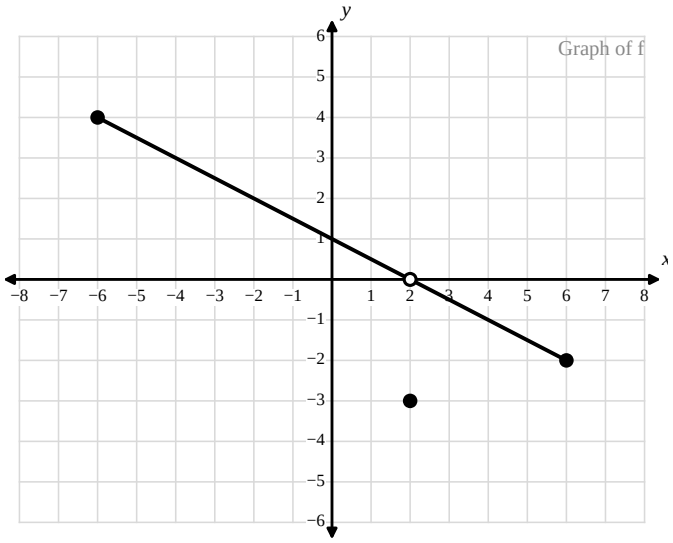


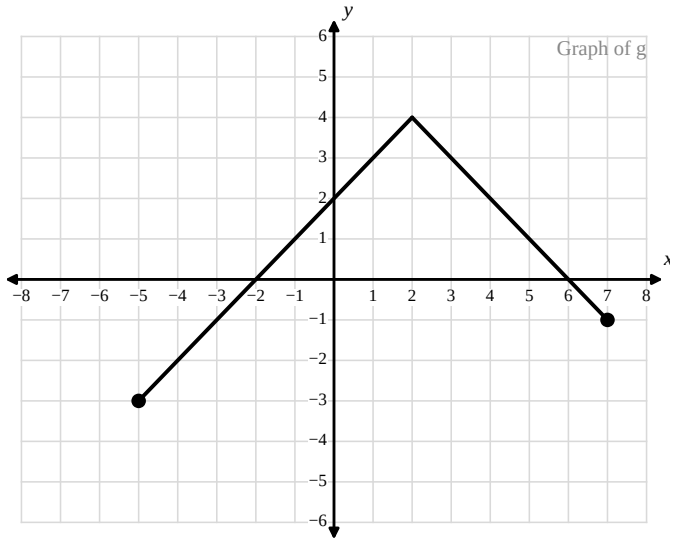

| | | |
|---|---|---|
| **1.** $\lim_{x\to 1} f(x) =$ | **2.** $\lim_{x\to 3} f(x) =$ | **3.** $\lim_{x\to 5} f(x) =$ |

**(2 points)**

**4.** Use the graph of $f$ above to determine all values of $a$ for which $\lim_{x\to a} f(x) = -1$ on the interval $-9 < x < 9$. If there are no values of $a$, write the empty set.

**Use the graphs of $f$ and $g$ below to find the limit. Show the limit properties you use. (2 points)**




**5.** $\lim_{x \to 4} [f(x)\, g(x)]$

**(2 points)**

**6.** Use the table to estimate $\lim_{x \to 2} h(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

| $x$ | 1.9 | 1.99 | 1.999 | $\to 2 \leftarrow$ | 2.001 | 2.01 | 2.1 |
|---|---|---|---|---|---|---|---|
| $h(x)$ | 11.41 | 11.9401 | 11.994001 | | 12.006001 | 12.0601 | 12.61 |

$\lim_{x \to 2^-} h(x) \approx$ _____ $\lim_{x \to 2^+} h(x) \approx$ _____ $\lim_{x \to 2} h(x) \approx$ _____

**Evaluate each limit. (2 points each)**

**7.** $\lim_{x \to 1} f(x), \quad f(x) = \begin{cases} x^2 + 1, & x < 1 \\ 3 - x, & x \geq 1 \end{cases}$

**8.** $\lim_{x \to 3} \dfrac{2x^2 - 5x - 3}{x - 3}$

**9.** $\lim_{x \to 4} \dfrac{3 - \sqrt{x + 5}}{x - 4}$

**10.** $\lim_{x \to 4} \dfrac{\frac{1}{x} - \frac{1}{4}}{x - 4}$

**11.** Find the average rate of change of $f(x) = \sqrt{x}$ from $4$ to $x$, then find its limit as $x \to 4$.

PCBC H Name **ANSWER KEY** **22** /22

# Limits Quiz 1.1–1.2

*Non-Calculator*

You have 20 minutes.

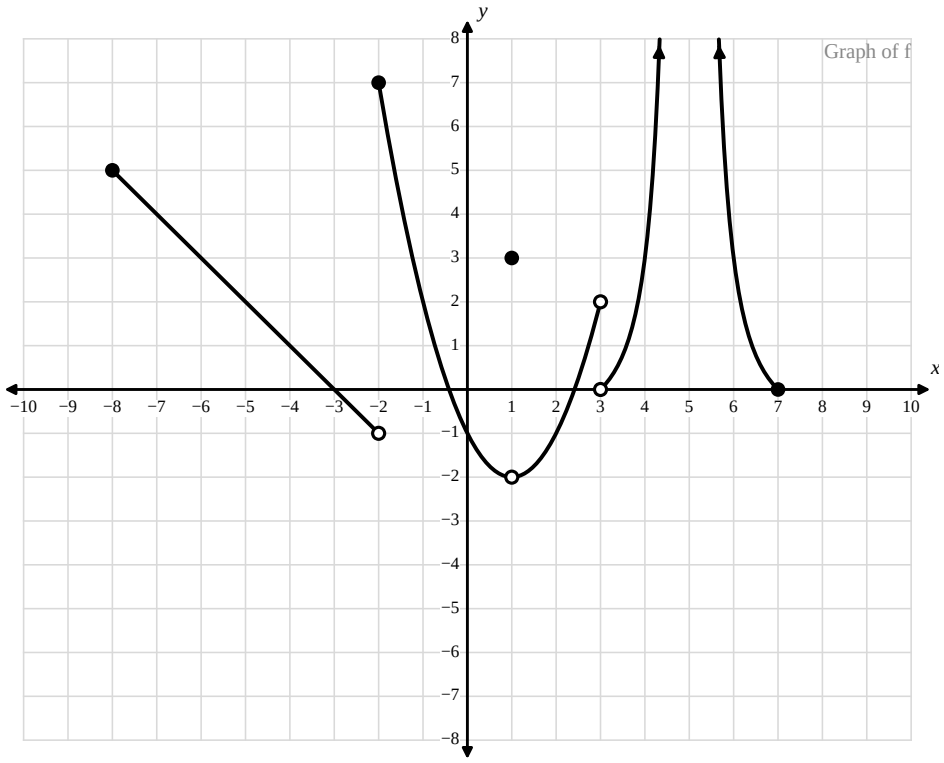
**PRACTICE C · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator**

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Signature _____

**SHOW ALL WORK FOR FULL CREDIT.**

**Use the graph of $f$ shown below to find each. If a limit does not exist, write DNE. (2 points each)**




**1.** $\lim_{x\to 1} f(x) =$

*Both sides of the parabola head to the open circle at the vertex* $(1, -2)$.

$-2$

**2.** $\lim_{x\to 3} f(x) =$

$\lim_{x\to 3^-} f(x) = 2$, $\lim_{x\to 3^+} f(x) = 0$

$2 \neq 0$, *so* $\lim_{x\to 3} f(x)$ *does not exist.*

DNE

**3.** $\lim_{x\to 5} f(x) =$

*Both sides of* $x = 5$ *rise without bound.*

$\infty$

**(2 points)**

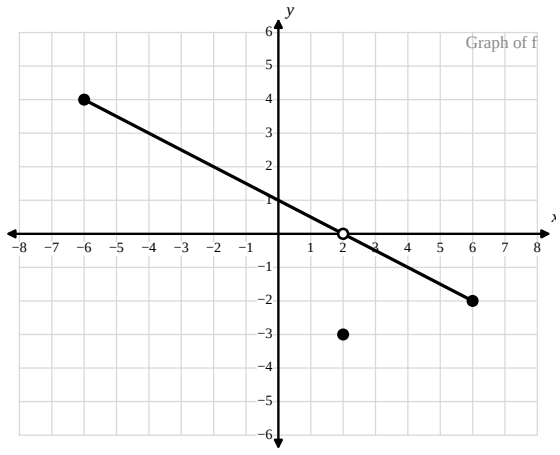
**4.** Use the graph of $f$ above to determine all values of $a$ for which $\lim_{x\to a} f(x) = -1$ on the interval $-9 < x < 9$. If there are no values of $a$, write the empty set.

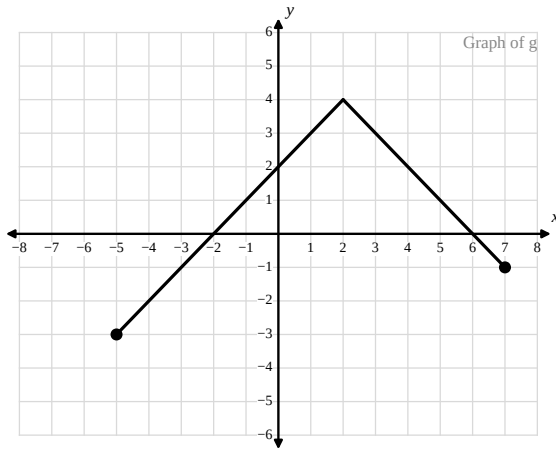
*Height* $-1$ *is reached from both sides at* $x = 0$ *and* $x = 2$.

*Not* $x = -2$*: only the left line heads to* $-1$*; the right side starts at 7.*

$a = 0,\ a = 2$

**Use the graphs of $f$ and $g$ below to find the limit. Show the limit properties you use. (2 points)**




**5.** $\lim_{x\to 4}[f(x)\,g(x)]$

$$\lim_{x\to 4}[f(x)g(x)] = \left[\lim_{x\to 4} f(x)\right]\left[\lim_{x\to 4} g(x)\right]$$
$$= (-1)(2) = -2$$

$-2$

**(2 points)**

**6.** Use the table to estimate $\lim_{x\to 2} h(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

| $x$ | 1.9 | 1.99 | 1.999 | $\to 2 \leftarrow$ | 2.001 | 2.01 | 2.1 |
|---|---|---|---|---|---|---|---|
| $h(x)$ | 11.41 | 11.9401 | 11.994001 | | 12.006001 | 12.0601 | 12.61 |

*Left: 11.41, 11.9401, 11.994001*
*Right: 12.61, 12.0601, 12.006001*
*Both sides agree. A table estimates; it does not prove.*

$\approx 12$

$\lim_{x\to 2^-} h(x) \approx$ 12 $\lim_{x\to 2^+} h(x) \approx$ 12 $\lim_{x\to 2} h(x) \approx$ 12

**Evaluate each limit. (2 points each)**

**7.** $\lim_{x\to 1} f(x), \quad f(x)=\begin{cases} x^2+1, & x<1 \\ 3-x, & x\geq 1 \end{cases}$

$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-}(x^2+1) = 1+1 = 2$

$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+}(3-x) = 3-1 = 2$

$\lim_{x\to 1} f(x) = 2$

$\boxed{2}$

**8.** $\lim_{x\to 3} \dfrac{2x^2-5x-3}{x-3}$

*Substitution produces 0/0: indeterminate.*

$$\lim_{x\to 3} \frac{2x^2-5x-3}{x-3} = \lim_{x\to 3} \frac{(2x+1)(x-3)}{x-3}$$
$$= \lim_{x\to 3}(2x+1) = 2(3)+1 = 7$$

$\boxed{7}$

**9.** $\lim_{x\to 4} \dfrac{3-\sqrt{x+5}}{x-4}$

*Substitution produces 0/0: indeterminate.*

$$\lim_{x\to 4} \frac{3-\sqrt{x+5}}{x-4} = \lim_{x\to 4} \frac{9-(x+5)}{(x-4)\left(3+\sqrt{x+5}\right)}$$
$$= \lim_{x\to 4} \frac{-(x-4)}{(x-4)\left(3+\sqrt{x+5}\right)}$$
$$= \lim_{x\to 4} \frac{-1}{3+\sqrt{x+5}} = \frac{-1}{3+3} = -\frac{1}{6}$$

$\boxed{-\frac{1}{6}}$

**10.** $\lim_{x\to 4} \dfrac{\frac{1}{x}-\frac{1}{4}}{x-4}$

*Substitution produces 0/0: indeterminate.*

$$\lim_{x\to 4} \frac{\frac{1}{x}-\frac{1}{4}}{x-4} = \lim_{x\to 4} \frac{4-x}{4x(x-4)}$$
$$= \lim_{x\to 4} \frac{-(x-4)}{4x(x-4)} = \lim_{x\to 4} \frac{-1}{4x}$$
$$= \frac{-1}{4(4)} = -\frac{1}{16}$$

$\boxed{-\frac{1}{16}}$

**11.** Find the average rate of change of $f(x)=\sqrt{x}$ from 4 to $x$, then find its limit as $x\to 4$.

$$\frac{f(x)-f(4)}{x-4} = \frac{\sqrt{x}-2}{x-4}$$

$$\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4} = \lim_{x\to 4} \frac{\sqrt{x}-2}{(\sqrt{x}-2)(\sqrt{x}+2)}$$
$$= \lim_{x\to 4} \frac{1}{\sqrt{x}+2} = \frac{1}{2+2} = \frac{1}{4}$$

$\boxed{\frac{\sqrt{x}-2}{x-4};\ \frac{1}{4}}$